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DATA ANALYST GRAPH ANALYSIS MASTER NOTES

(Company Level | Interview | Projects | Business Insights)

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★ DATA ANALYST KA ASLI KAAM

Bahut log sochte hain Data Analyst ka kaam graph banana hai.

✗ Galat

Data Analyst ka asli kaam hai:

Data



Graph



Observation



Business Insight



Recommendation



Business Decision

Company graph ke paise nahi deti.

Company Decision ke paise deti hai.

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🔥 GOLDEN GRAPH ANALYSIS FORMULA

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Har graph ko isi order me analyse karo.

STEP 1

Graph ka Title padho

Question:

Ye graph kis topic ke baare me hai?

STEP 2

X-Axis dekho

Question:

X-Axis me kya hai?

Examples

Months

Products

Cities

Employees

Years

Customers

STEP 3

Y-Axis dekho

Question:

Y-Axis me kya measure ho raha hai?

Examples

Sales

Profit

Revenue

Salary

Population

Count

Price

STEP 4

Highest Value dhundo

Question

Sabse highest value kiski hai?

Highest value generally

Best Product

Best City

Best Employee

Best Department

Highest Revenue

Highest Sales

ko represent karti hai.

STEP 5

Lowest Value dhundo

Question

Sabse lowest value kiski hai?

Lowest value generally

Weak Product

Weak Sales

Weak Department

Low Performance

ko represent karti hai.

STEP 6

Pattern dekho

Question

Kya graph me koi pattern dikh raha hai?

Examples

Increasing

Decreasing

Stable

Random

Seasonal

STEP 7

Outlier dhundo

Question

Kya koi unusual value hai?

Example

Salary

₹20,000

₹22,000

₹25,000

₹30,000

₹8,00,000

₹8,00,000

Outlier.

Check

CEO

ya

Data Entry Error

STEP 8

Trend dekho

Question

Time ke saath graph upar ja raha hai?

Ya niche?

Ya stable hai?

STEP 9

Observation likho

Observation

Sirf graph me jo dikh raha hai wahi likho.

Example

TV category ki sales sabse zyada hai.

Laptop category ki sales sabse kam hai.

Revenue continuously increase ho raha hai.

Customer age mostly 20-30 hai.

Bas.

Guess mat karo.

STEP 10

Reason socho

Question

Ye kyu hua?

Possible Reasons

Marketing

Festival

Discount

Brand Value

Competition

Price

Quality

Customer Demand

STEP 11

Business Impact

Question

Iska company par kya effect padega?

Example

Revenue Increase

Profit Increase

Customer Loss

Inventory Issue

High Demand

Low Demand

Market Growth

STEP 12

Recommendation

Question

Manager ko kya suggest karoge?

Examples

Stock Increase

Marketing Increase

Discount Launch

Price Reduce

More Hiring

Inventory Improve

Customer Survey

Training Program

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🔥 GRAPH ANALYSIS TEMPLATE

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Graph Name

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Title

↓

X-Axis



Y-Axis



Highest Value



Lowest Value



Pattern



Outlier



Observation



Possible Reason



Business Impact



Recommendation

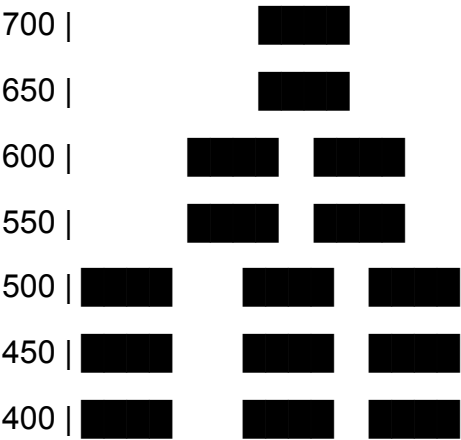
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EXAMPLE 1

BAR CHART

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Sales



Laptop Phone TV

Analysis

Title

Monthly Product Sales

X-Axis

Products

Y-Axis

Sales

Highest

TV

Lowest

Laptop

Pattern

TV > Phone > Laptop

Observation

TV category generated maximum sales.

Laptop category generated minimum sales.

Possible Reason

High demand

Better marketing

Customer preference

Business Impact

High revenue from TV.

Low revenue from Laptop.

Recommendation

Increase TV stock.

Increase TV marketing.

Launch offers on Laptop.

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EXAMPLE 2

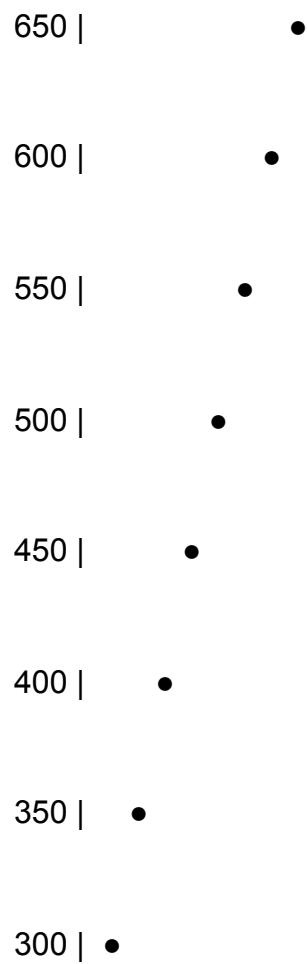
LINE CHART

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Revenue

700 |





Jan Feb Mar Apr May Jun Jul Aug

Analysis

Observation

Revenue is increasing every month.

Possible Reason

Marketing successful.

More customers.

Good product demand.

Business Impact

Company revenue increasing.

Business growth healthy.

Recommendation

Increase production.

Increase inventory.

Continue marketing.

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EXAMPLE 3

SCATTER PLOT

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Salary

80k | ●

70k |

60k | •

50k |

40k | •

30k | •

20k |

Experience

Analysis

Observation

Salary increases with experience.

Possible Reason

Experienced employees receive higher salary.

Business Impact

Experienced employees are valuable.

Recommendation

Employee retention.

Training Program.

Career Growth Plan.

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EXAMPLE 4

HEATMAP

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Salary

Age

Experience

Salary

1.00

0.65

0.94

Age

0.65

1.00

0.55

Experience

0.94

0.55

1.00

Observation

Salary and Experience have strong relationship.

Age has weak relationship.

Business Impact

Experience is important feature.

Recommendation

Use Experience for prediction.

Ignore unnecessary features.

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EXAMPLE 5

HISTOGRAM

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Customers

40 |

35 | ■

30 | ■■■■■

25 | ■■■■■■■■■■

20 | ■■■■■■■■■■

18-25

25-35

35-45

Observation

Most customers belong to age group 18-25.

Possible Reason

Young audience interested.

Business Impact

Young customers generate majority sales.

Recommendation

Target Instagram Ads.

Student Offers.

Youth Marketing.

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EXAMPLE 6

BOX PLOT

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20k

22k

25k

27k

30k

↓

₹10,00,000

Observation

One salary is extremely high.

Possible Reason

CEO

Manager

Data Error

Business Impact

Average salary may become incorrect.

Recommendation

Verify Outlier.

Remove only if incorrect.

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EXAMPLE 7

PIE CHART

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Apple

45%

Samsung

35%

Xiaomi

20%

Observation

Apple has highest market share.

Business Impact

Apple dominates market.

Recommendation

Study competitor strategy.

Improve product marketing.

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BUSINESS INSIGHT WRITING FORMULA

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Observation

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Reason

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Business Impact

↓

Recommendation

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INTERVIEW ANSWER FORMULA

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1. Graph Name

2. Purpose

3. Observation

4. Business Insight

5. Recommendation

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5 GOLDEN QUESTIONS

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1. Graph kya dikha raha hai?

2. Highest value kiski hai?

3. Lowest value kiski hai?

4. Ye kyu hua hoga?

5. Company ko ab kya action lena chahiye?

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🔥 DATA ANALYST MINDSET

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Student bolta hai:

"TV ki sales zyada hai."

Data Analyst bolta hai:

"TV category generated the highest sales, indicating strong customer demand. The company should increase inventory, continue marketing investment, and ensure stock availability to maximise revenue."

Yahi difference hai Student aur Professional Data Analyst me.

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